

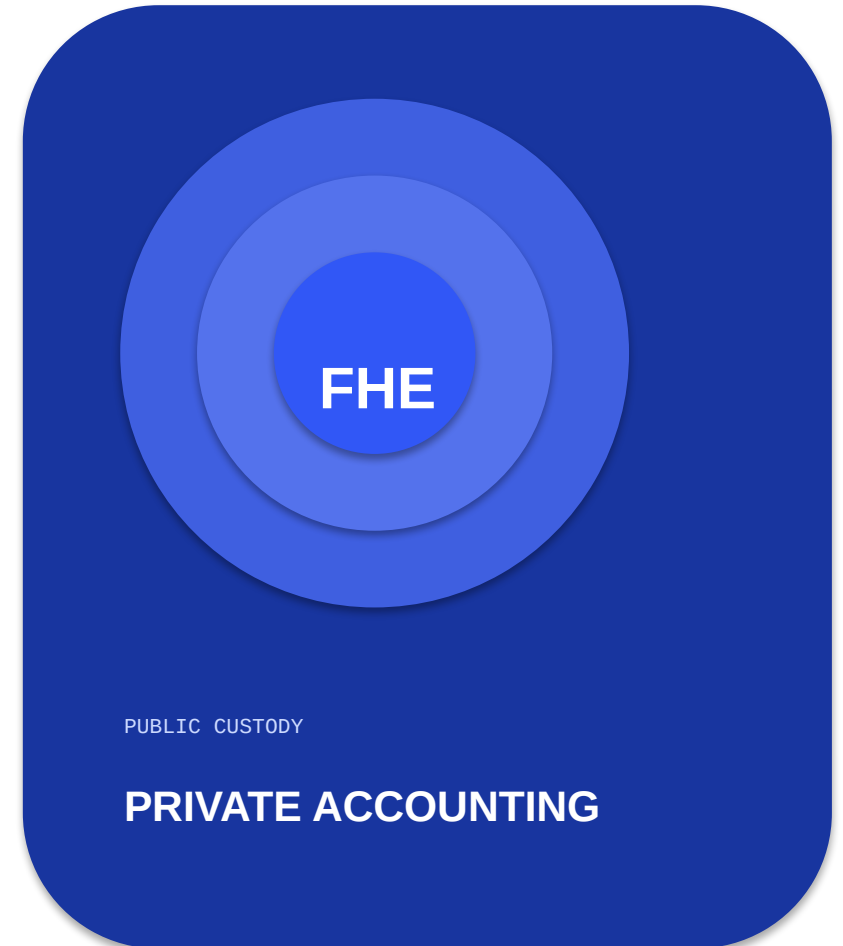
ZAMA FHEVM · CONFIDENTIAL FINANCE

Private prize savings, built for verification.

Principal stays withdrawable while encrypted balances and ticket weights stay confidential.

cipherpool-beta.vercel.app

RESEARCH SOFTWARE · NOT PRODUCTION



THE PROBLEM

Public ledgers expose private financial context

Auditability should not require publishing every saver's position.

01

Balance exposure

Wallet history can reveal a saver's position, deposits, withdrawals, and changing financial behavior.

02

Visible ticket weights

Transparent weights make whale activity easy to track and can disclose each participant's relative odds.

03

Timing reveals intent

Pending deposits and last-minute entries expose behavior around a prize round before it settles.

THE SOLUTION

Public custody. Confidential accounting.

CipherPool separates what must be auditable from what should remain private.

PUBLIC · INDEPENDENTLY VERIFIABLE

Custody layer

- Deposited testnet cUSDC
- Contract and custody bindings
- Transaction receipts
- Aggregate solvency

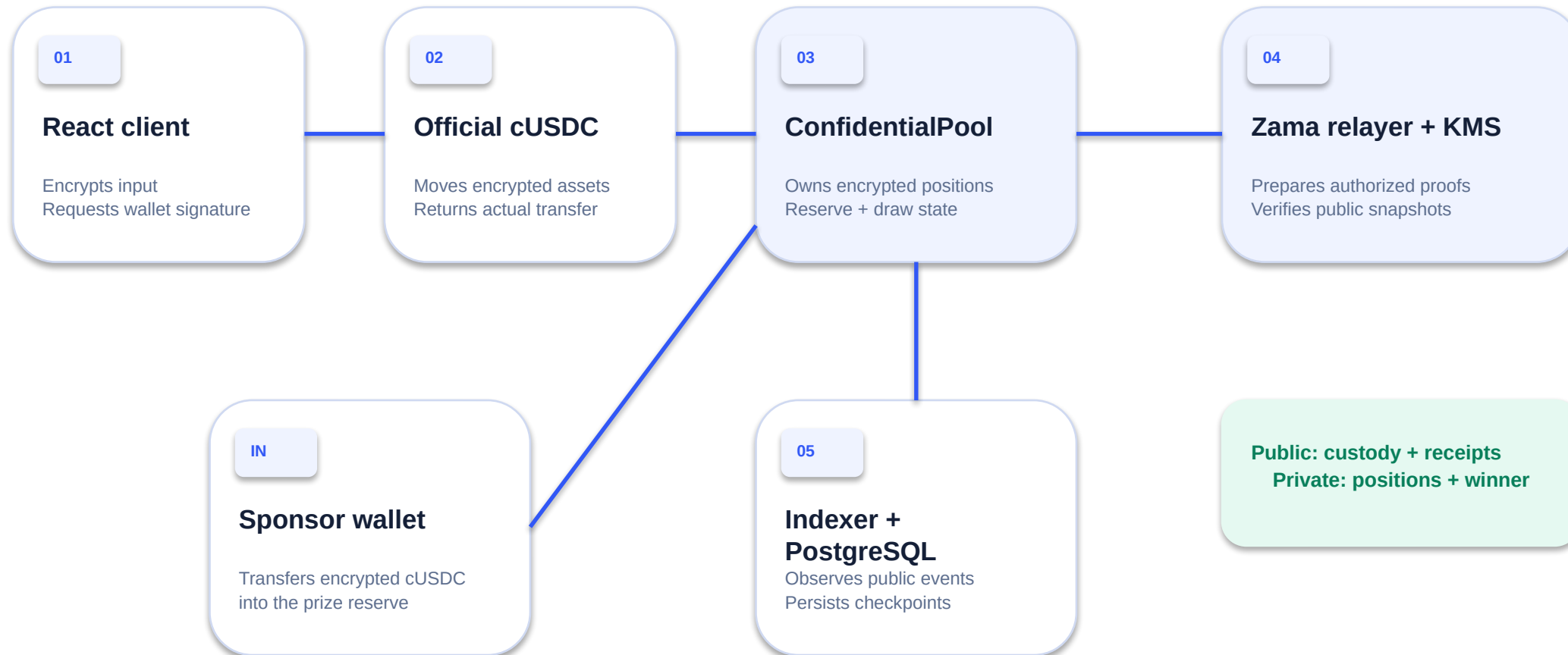
ENCRYPTED · WALLET-AUTHORIZED

Privacy layer

- Running balances as euint64
- Weighted draw intervals
- Prize credits
- Local balance reveal

ARCHITECTURE

A verifiable path from wallet to confidential settlement



LIVE PRODUCT

The console is live—and honest about connection state

 **CipherPool**
PRIVATE PRIZE SAVINGS

[Save privately](#) [Prize rounds](#) [How it works](#)

[Connect wallet](#)

CONFIDENTIAL SAVINGS CONSOLE

Your position stays private.

Deposit confidential cUSDC, enter verifiable prize rounds, and withdraw without publishing amounts, balances, or ticket weight.

[Encrypted accounting](#) [Wallet-bound access](#) [Verifiable exits](#)

RUNTIME ASSURANCE **PENDING**

Protocol state

Network ● Wallet Disconnected

Indexer ● Online

Transactions ● Safety-Locked

⚠️ Writes remain locked until runtime verification succeeds. Connect a Sepolia wallet to verify the active deployment.

● Indexer Online ● Pool Checking ● Wallet Disconnected ● Deployment Pending

Checked 6:54:15 PM

VERIFIED POOL SNAPSHOT **STALE**
1 cUSDCMock
Last confirmed value · Aggregate only; individual positions stay encrypted

VERIFIED PRIZE RESERVE **STALE**
0 cUSDCMock
Last confirmed value · Sponsor-funded on Sepolia; last KMS-verified snapshot

PRIVATE SAVERS **LOADING**
—
Loading verified source...

CONFIRMED ROUNDS **LIVE**
1
Recorded on Sepolia

PRIVATE OPERATIONS

Manage your savings

Balances reveal only after wallet authorization. Deposits and settlement remain independently verifiable on-chain.

PRIVATE POSITION **ACL PROTECTED**
Only you can reveal this balance
The ciphertext stays on-chain.

STEP 01 **ENCRYPTED END-TO-END**
Fund your private position
The amount transfers as

STEP 02 **ONE TRANSACTION**
Withdraw confidentially
The requested amount and

PRODUCTION CAPTURE

Clear wallet state

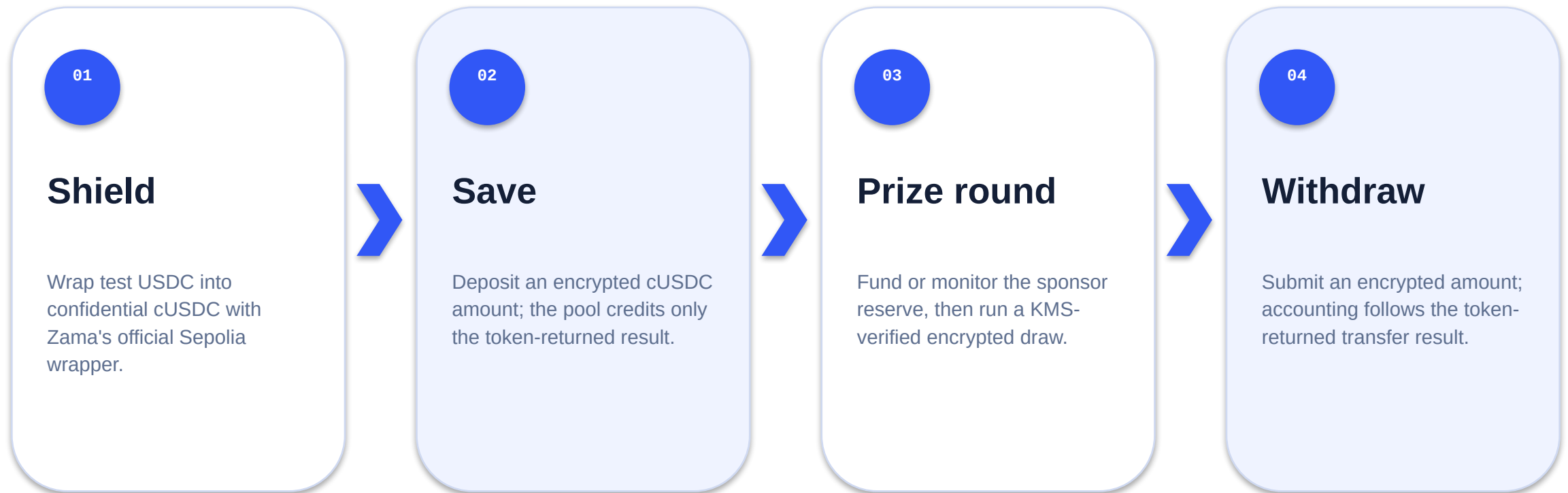
A disconnected user sees only Connect wallet. An address appears only after the provider returns the selected account.

Runtime assurance

Chain, bytecode, and custody bindings are checked before protocol writes.

Captured from <https://cipherpool-beta.vercel.app> · 4 Sep 2026

Four explicit steps. No hidden signing.



SECURITY CONTROLS

Solvency and settlement invariants are enforced in code

01

Asset-bound credit

Encrypted deposit credit is derived from the token-returned custody amount.

02

Reserved prizes

Each draw consumes verified sponsor funds so assets cannot fund repeated awards.

03

Complete liabilities

Winner credit enters both the encrypted position and aggregate at finalization.

04

Bound draw proofs

KMS evidence must match the stored aggregate and reserve handles.

05

Timeout recovery

Anyone can release a stale draw lock after the 24-hour cancellation delay.

VERIFIED EVIDENCE

A complete live prize round—not a simulated receipt

Each transaction below links directly to its confirmed block-explorer record.

01

cUSDC deposit

[0xe36db7...b39f](#)

Encrypted position

02

KMS draw

[0x504862...ce6c](#)

Draw 1 finalized

03

Private claim

[0x5763be...c969](#)

Ordinary withdrawal

Deposit → KMS draw → private claim → principal withdrawal · all confirmed

Reproducible tests—and durable live infrastructure

FULL

VALIDATION SUITE

Foundry contracts · backend API/indexer
client encryption · frontend UX

Vercel

React frontend

PRODUCTION UI

Render

Node indexer/API

PUBLIC API

Neon

PostgreSQL checkpoint

RESTART RECOVERY

Indexer checkpoint restored successfully after a live service restart

OPEN RESEARCH · VERIFIABLE EVIDENCE

Confidential savings, without blind trust.

Try the live app. Inspect the active Sepolia deployment. Review every invariant and test.

cipherpool-beta.vercel.app

github.com/Webghost01-NG/fhevm-pooltogether-security

ACTIVE POOL

0x9c939b82
a1B23b77
746f934A
1Ff2b9a5
bCf191e0

ETHEREUM SEPOLIA